Three EEG measures, three temporal anchors

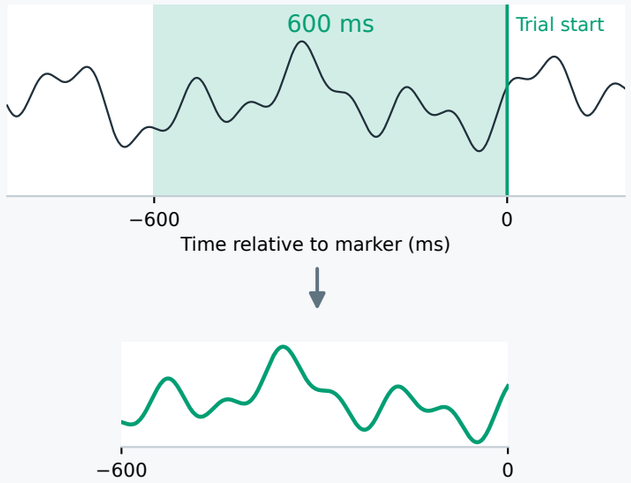
Preprocessed continuous EEG → trial-level time courses → a common representational analysis

Fixation cross → **Trial starts → encoding** → **Full sequence + choices** → **Response**

Trial phases shown schematically; durations are not to scale.

Pre-trial rest EEG

Before the trial-start marker

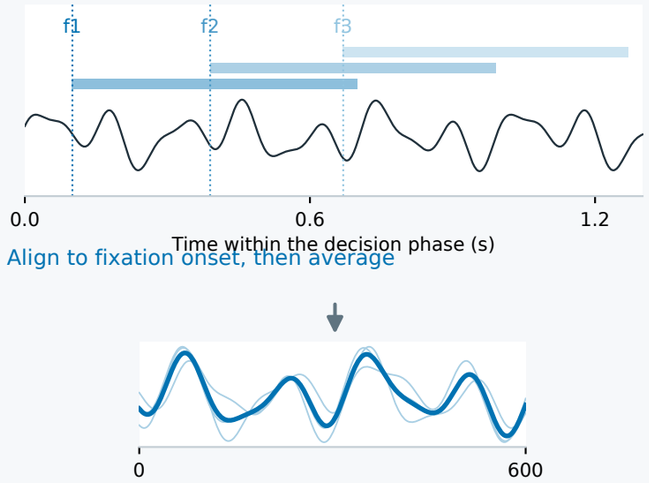


One pre-trial segment per trial

Fixation cross displayed before trial onset

Fixation-related EEG (FRP)

After each retained sequence fixation

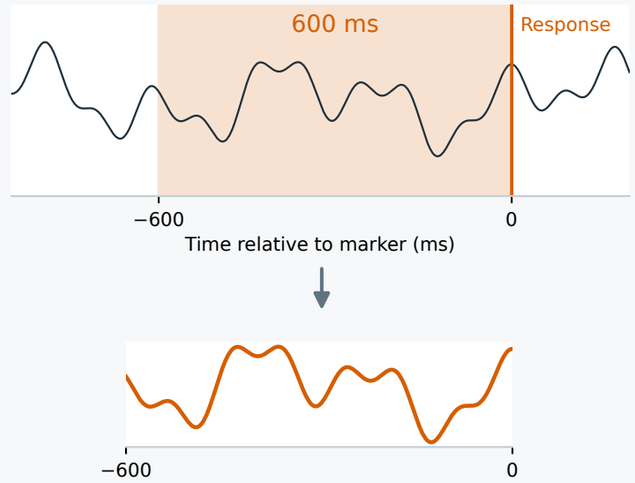


One trial FRP from multiple epochs

Sequence fixations ≥100 ms; windows may overlap

Response-locked EEG

Before the response marker



One pre-response segment per trial

Omissions: end of the response window

Shared construction of representational geometry

Regional time courses
Average electrodes
at each time point

→

8 pattern vectors
Average trials
within pattern + session

→

Session RDM
Pairwise correlation distance:
1 − r

→

Participant → group
Average RDMs
within, then across people

Repeat separately for each EEG measure and each of five electrode groups: frontal, temporal, central, occipital and parietal.

Same 600-ms duration does not imply equivalent measurements.

FRP averages multiple post-fixation windows; response and rest use one window per trial. No fixation-specific baseline correction is applied to FRPs.

Rest denotes the pre-trial fixation-cross interval, not a separate resting-state recording. Waveforms and timing are schematic, not measured data.